The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 June 2021.

INCH-POUND

MIL-PRF-19500/555M
w/AMENDMENT 2
8 March 2021
SUPERSEDING
MIL-PRF-19500/555M
w/AMENDMENT 1
18 May 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT, N-CHANNEL, SILICON, TYPES 2N6788, 2N6790, 2N6792, AND 2N6794, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode, MOSFET, power transistor. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device.

1.2 Package outlines. The device package outlines are as follows: TO-205AF in accordance with figure 1 and leadless chip carrier (LCC) in accordance with figure 2 for all encapsulated device types. See figures 3, 4, 5, 6, and 7 for all unencapsulated devices.

1.3 Maximum ratings. (Unless otherwise specified, T_A = +25°C).

Type	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	V_{DS} and V_{DG} 70,000 ft. altitude
	W	V dc	V dc	V dc	
2N6788, 2N6788U	0.8	100	100	±20	
2N6790, 2N6790U	0.8	200	200	±20	
2N6792, 2N6792U	0.8	400	400	±20	300
2N6794, 2N6794U	0.8	500	500	±20	300

Type	P_T (1) T_C = +25°C	$R_{\theta JC}$ (2)	I_{D1} (3) (4) T_C = +25°C	I_{D2} T_C = +100°C	I_S	I_{DM} (5)	T_J and T_{STG}
	W	°C/W	A dc	A dc	A dc	A (pk)	°C
2N6788	20	6.25	6.0	3.5	6.0	24	-55°
2N6790	20	6.25	3.5	2.25	3.5	14	to
2N6792	20	6.25	2.0	1.25	2.0	8	+150°
2N6794	20	6.25	1.5	1.0	1.5	6	
2N6788U	14	8.93	4.5	2.8	4.5	18	
2N6790U	14	8.93	2.8	1.8	2.8	11	
2N6792U	14	8.93	1.8	1.13	1.8	7.2	
2N6794U	14	8.93	1.4	0.88	1.4	5.6	

See notes on next page.

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1.3 Maximum ratings. Continued.

(1) Derate linearly 0.16 W/°C for T_C > +25°C for "non-U" suffix versions, 0.11 W/°C for T_C > +25°C for "U" suffix versions.
(2) See figure 8, thermal impedance curves.
(3) The following formula derives the maximum theoretical I_D limit. I_D is also limited by package and internal wires and may be limited due to pin diameter.

$$I_D = \sqrt{\frac{T_J(\max) - T_C}{R_{\theta JC} x (R_{DS(on)} at T_J(\max))}}$$

(4) See figure 9, maximum drain current graph.
(5) I_{DM} = 4 I_{D1}; I_{D1} as calculated in footnote (3).

1.4 Unless otherwise specified, primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0, V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)}$ (1) V_{GS} = 10 V dc I_D = I_{D2}	
					T_J = +25°C	T_J = +150°C
	V dc	V dc		µA dc	Ohms	Ohms
		Min	Max			
2N6788, 2N6788U	100	2.0	4.0	25	0.30	0.60
2N6790, 2N6790U	200	2.0	4.0	25	0.80	1.80
2N6792, 2N6792U	400	2.0	4.0	25	1.80	4.50
2N6794, 2N6794U	500	2.0	4.0	25	3.00	7.50

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.5.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "6788", "6790", “6792”, and "6794".

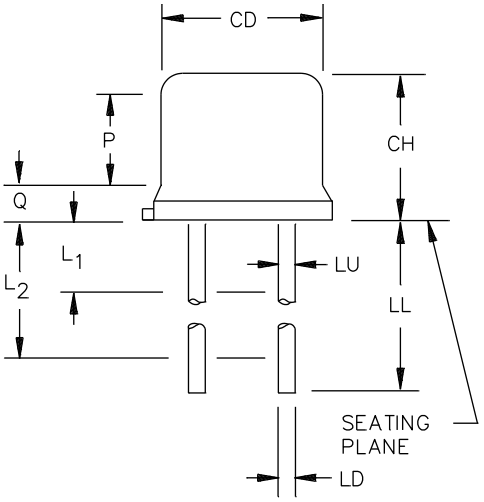
1.5.3.3 Suffix letters. No suffix letters are used on devices that are packaged in the TO-205AF package of figure 1. The suffix letter "U" is used on devices that are packaged in the LCC package of figure 2.

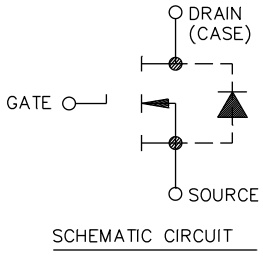
1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

* 1.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A", “C”, “D”, "E", and "F" (see figures 3, 4, 5, 6, and 7).

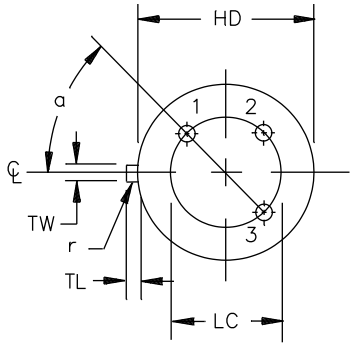
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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.315	.355	8.01	9.01	
CH	.160	.180	4.07	4.57	
HD	.340	.370	8.64	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.019	0.41	0.48	7,8
LL	.500	.750	12.70	19.05	7,8,12
LU	.016	.019	0.41	0.48	7,8
L1		.050		1.27	7,8
L2	.250		6.35		7,8
P	.100		2.54		
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	3,4
TW	.028	.034	0.72	0.86	3
r		.010		0.25	10
α	45° TP		45° TP		6





SCHEMATIC CIRCUIT



NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of .011 inch (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.

* 11. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

12. Lead 1 = source, lead 2 = gate, lead 3 = drain.

FIGURE 1. Physical dimensions (similar to TO-205AF) 2N6788, 2N6790, 2N6792, and 2N6794.

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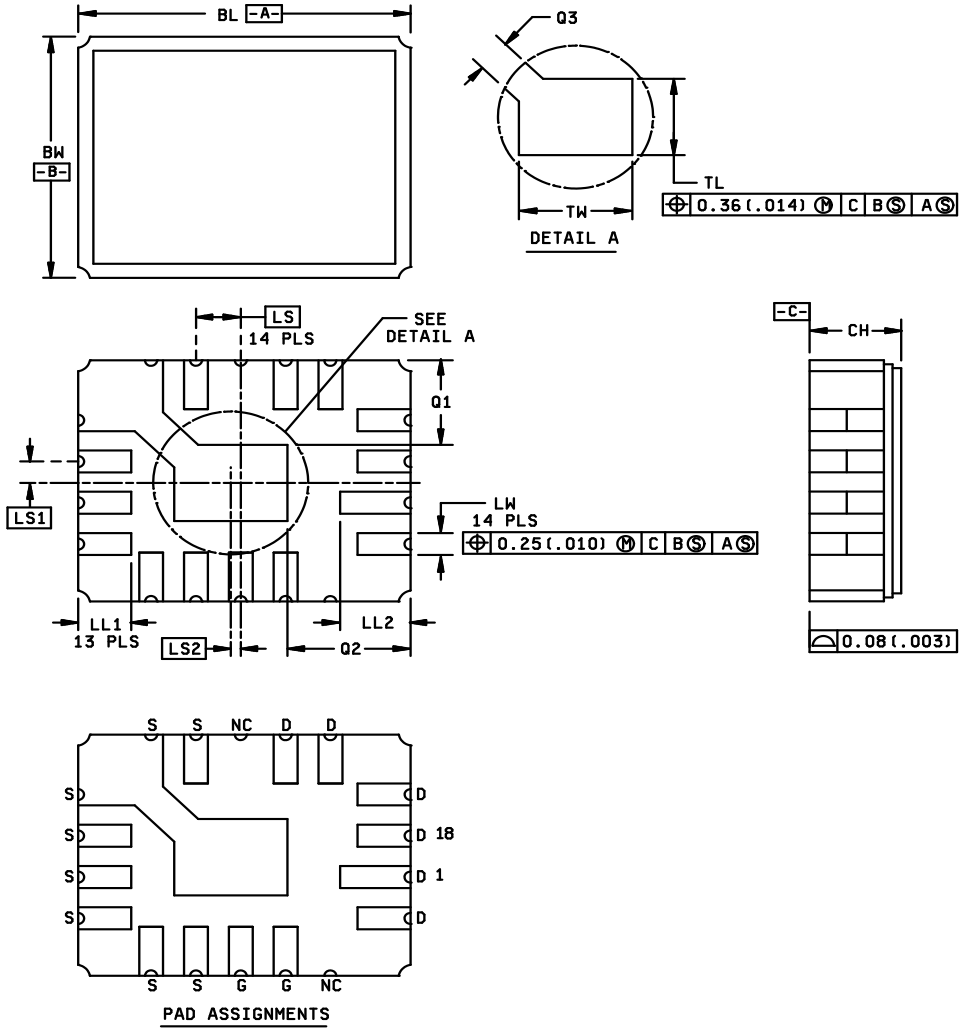


FIGURE 2. Physical dimensions for LCC (2N6788U, 2N6790U, 2N6792U, and 2N6794U).

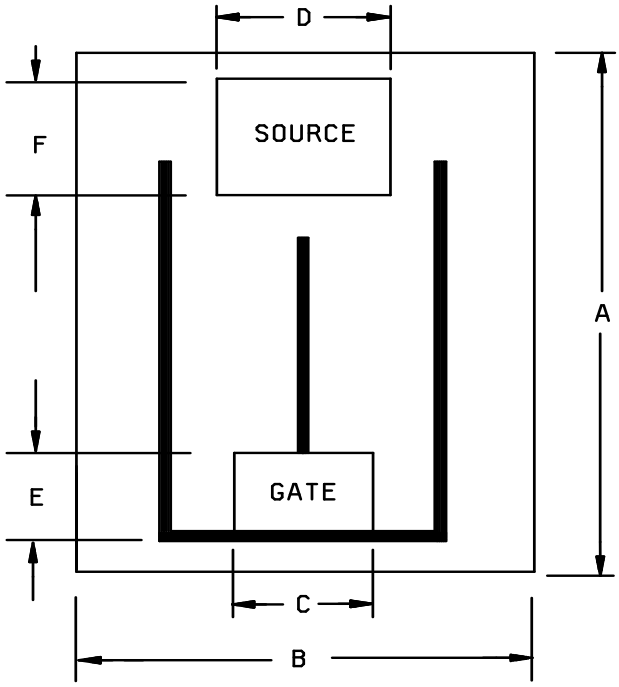
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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.345	.360	8.77	9.14
BW	.280	.295	7.12	7.49
CH	.095	.115	2.42	2.92
LL_1	.040	.055	1.02	1.39
LL_2	.055	.065	1.40	1.65
LS	.050 BSC		1.27 BSC	
LS_1	.025 BSC		0.635 BSC	
LS_2	.008 BSC		0.203 BSC	
LW	.020	.030	0.51	0.76
Q_1	.105 REF		2.67 REF	
Q_2	.120 REF		3.05 REF	
Q_3	.045	.055	1.15	1.39
TL	.070	.080	1.78	2.03
TW	.120	.130	3.05	3.30

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
* 3. Dimensions and tolerancing shall be in accordance with ASME Y14.5.

FIGURE 2. Physical dimensions for LCC (2N6788U, 2N6790U, 2N6792U, and 2N6794U) - Continued.

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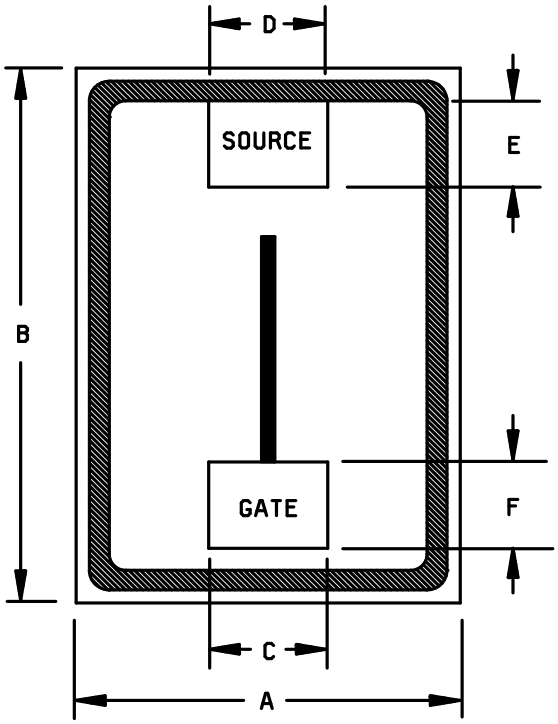
A version

Ltr	Dimensions - 2N6788				Dimensions - 2N6790				Dimensions - 2N6792, 2N6794			
	Inches		Millimeters		Inches		Millimeters		Inches		Millimeters	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
A	.1043	.120	2.649	3.049	.094	.112	2.39	2.85	.131	.147	3.33	3.73
B	.0717	.0874	1.820	2.220	.083	.099	2.11	2.52	.090	.106	2.29	2.69
C	.0245	.0249	0.623	0.633	.018	.028	0.46	0.71	.022	.032	0.56	0.81
D	.0322	.0326	0.817	0.827	.028	.038	0.71	0.97	.028	.038	0.71	0.97
E	.0177	.0181	0.449	0.459	.015	.025	0.38	0.64	.015	.025	0.38	0.64
F	.0238	.0242	0.604	0.614	.018	.028	0.46	0.71	.020	.030	0.51	0.76

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Die thickness = .019 ± .005 inch (0.48 ± 0.13 mm).
4. Back metal: Cr - Ni - Ag.
5. Top metal: Al.
6. Back contact: Drain.
7. Layout of gate fingers shown is typical, specific layout register in accordance with DSCC Form 36D.

* FIGURE 3. Physical dimensions JANHCA and JANKCA.

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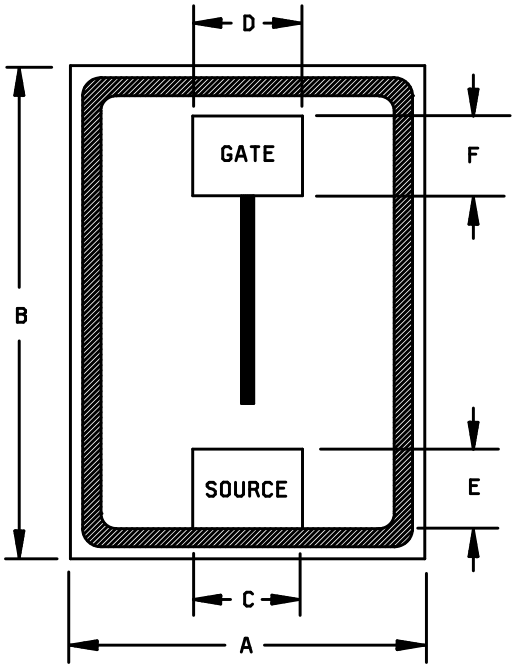
C version

Ltr	Dimensions 2N6788, 2N6790			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.0858	.0898	2.18	2.28
B	.087	.091	2.21	2.31
C	.0258	.0298	0.65	0.76
D	.0253	.0293	0.64	0.74
E	.017	.021	0.43	0.53
F	.016	.020	0.41	0.51

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Die thickness = .015 ±.005 inch (0.38 ±0.13 mm).
4. Back metal: Ag - Ti - Ni.
5. Top metal: Al.
6. Back contact: Drain.

FIGURE 4. Physical dimensions JANHCC and JANKCC.

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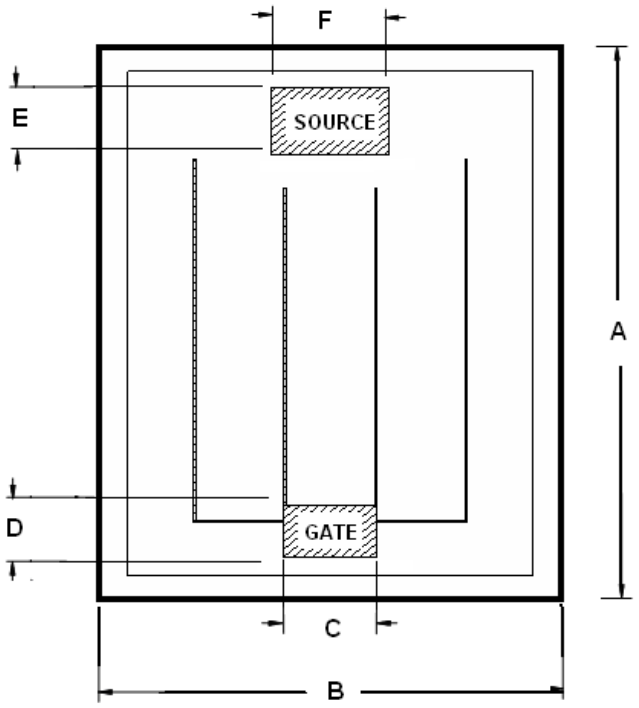
D version

Ltr	Dimensions 2N6792, 2N6794			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.093	.102	2.36	2.59
B	.140	.144	3.56	3.66
C	.025	.029	0.64	0.74
D	.026	.030	0.66	0.76
E	.016	.020	0.41	0.51
F	.017	.021	0.43	0.53

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Die thickness = .015 ±.005 inch (0.38 ±0.13 mm).
4. Back metal: Ag - Ti - Ni.
5. Top metal: Al.
6. Back contact: Drain.

FIGURE 5. Physical dimensions JANHCD and JANKCD.

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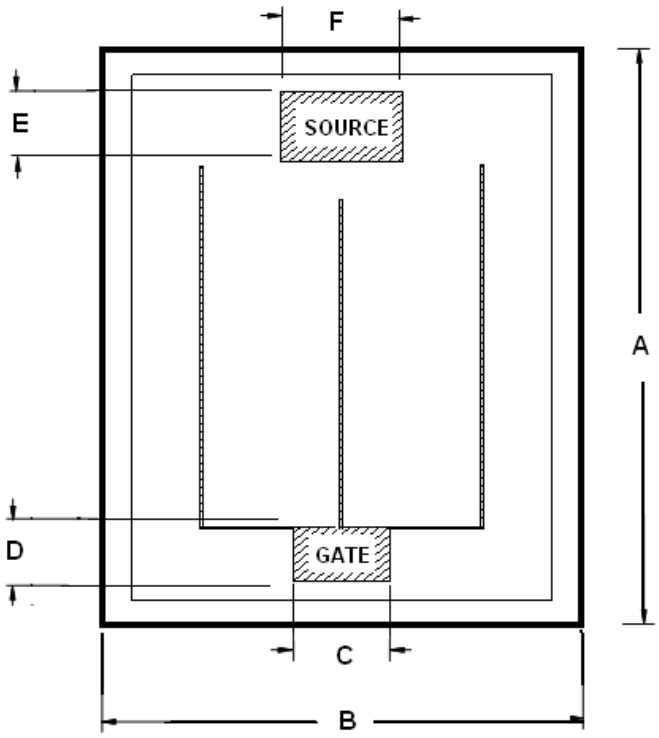


Ltr	Dimensions - 2N6788			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.093	.097	2.36	2.46
B	.123	.127	3.12	3.22
C	.021	.025	.53	.63
D	.013	.017	.33	.43
E	.014	.018	.36	.46
F	.023	.027	.58	.69

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .015 inch (0.38 mm) ±.001 inch (0.025 mm).

FIGURE 6. JANHCE and JANKCE (E-version) die dimensions for 2N6788.

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Ltr	Dimensions - 2N6790			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.091	.095	2.31	2.41
B	.123	.127	3.12	3.22
C	.021	.025	.53	.63
D	.011	.015	.28	.38
E	.013	.017	.33	.43
F	.023	.027	.58	.69

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .015 inch (0.38 mm) ±.001 inch (0.025 mm).

FIGURE 7. JANHCF and JANKCF (F-version) die dimensions for 2N6790.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.
* MIL-STD-883 - Test Method Standard for Microcircuits.

* (Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

I_{AS} - Rated avalanche current, nonrepetitive.
nC - nano coulomb.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (TO-205AF), 2 (LCC), and figures 3, 4, 5, 6, and 7 for JANHC and JANKC (die) herein.

3.4.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core is permitted (for TO-205AF). Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

* 3.4.3 Silicone die coating. The use of a silicone die coat requires a successful completion of MIL-STD-883, method 5011 on each silicone lot for its intended applications, and as part of the full MIL-PRF-19500 qualification process.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care should be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be as specified in MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed by the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500) (1) (2)	Measurement	
	JANS level	JANTX and JANTXV levels
(3)	Gate stress test (see 4.3.2)	Gate stress test (see 4.3.2)
(3) (4)	Method 3470 of MIL-STD-750 (see 4.3.3) optional	Method 3470 of MIL-STD-750 (see 4.3.3) optional
(3) 3c	Method 3161 of MIL-STD-750 (see 4.3.4)	Method 3161 of MIL-STD-750 (see 4.3.4)
* 5	Method 2052 of MIL-STD-750, PIND (see MIL-PRF-19500 and 4.3.5)	Not applicable
9	Subgroup 2 of table I herein	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein,
12	Method 1042 of MIL-STD-750, test condition A	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$= ±20 percent of initial value.	Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$= ±20 percent of initial value.

(1) At the end of the test program, I_{GSSF1} I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.
(4) Method 3470 is optional if performed as a sample in group A, subgroup 5.

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4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-PRF-19500, as a minimum, die shall be 100 percent probed in accordance with table I, subgroup 2, except test current shall not exceed 20 A.

4.3.2 Gate stress test. Apply V_{GS} = +30 V minimum for t = 250 µs minimum.

4.3.3 Single pulsed unclamped inductive switching.

a. Peak current, I_D ..... 2.2 A.

b. Peak gate voltage, V_{GS} ..... 10 V.

c. Gate to source resistor, R_{GS} ..... $25 \leq R_g \leq 200\Omega$.

d. Initial case temperature ..... +25°C, +10°C, -5°C.

e. Inductance, L ..... 100 µH (minimum)

f. Number of pulses to be applied ..... 1 pulse.

* 4.3.4 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). See table II, group E, subgroup 4 herein.

* 4.3.5 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and shall be evaluated for root cause and lot integrity.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for conformance inspection in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows.

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4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

	Subgroup	Method	Conditions
	B3	1051	Test condition G.
	B3	2037	Test condition D. All internal wires for each device shall be pulled separately.
*	B3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	B4	1042	Test condition D. The heating cycle shall be 1 minute minimum. No heat sink or forced air cooling on the device shall be permitted during the "on" cycle.
	B5	1042	A separate sample may be pulled for each test. Accelerated steady-state reverse bias; test condition A, V_{DS} = rated, T_A = +175°C, t = 120 hours, read and record $V_{BR(DSS)}$ (pre and post) at I_D = -1 mA. Read and record I_{DSS} (pre and post).
	B5	1042	Accelerated steady-state gate stress; test condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condition G.
B3	1042	Test condition D. The heating cycle shall be 1 minute minimum. No heat sink or forced air cooling on the device shall be permitted during the "on" cycle.
B3	2037	Test condition D. All internal wires for each device shall be pulled separately. If group B3 is to be continued to C6, bond strength test may be performed after C6.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

	Subgroup	Method	Conditions
	C2	2036	Test condition E (not required for LCC).
*	C3	2052	PIND, required if not performed in screening. (JANS only, 22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	C5	3161	See 4.3.4, for $R_{\theta JC}$ = 6.25°C/W for 2N6788, 2N6790, 2N6792, and 2N6794 $R_{\theta JC}$ = 8.93°C/W for 2N6788U, 2N6790U, 2N6792U, and 2N6794U
	C6	1042	Test condition D. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle (for JANS level devices with .008 inch or larger bond wires, that have completed 6,000 cycles in group B subgroup 4, C6 is not applicable).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.4	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1 mA dc, condition C	$V_{(BR)DSS}$			
2N6788, 2N6788U				100		V dc
2N6790, 2N6790U				200		V dc
2N6792, 2N6792U				400		V dc
2N6794, 2N6794U				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = + 20 V dc, V_{DS} = 0, bias condition C	I_{GSSF1}		+100	nA dc
Gate current	3411	V_{GS} = - 20 V dc, V_{DS} = 0, bias condition C	I_{GSSR1}		- 100	nA dc
Drain current	3413	V_{GS} = 0 V dc, V_{DS} = 80 percent of rated V_{DS}; bias condition C	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); bias condition A,	$r_{DS(on)1}$			
2N6788, 2N6788U		I_D = rated I_{D2} (see 1.3)			0.30	Ω
2N6790, 2N6790U					0.8	Ω
2N6792, 2N6792U					1.8	Ω
2N6794, 2N6794U					3.0	Ω
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); bias condition A,	$r_{DS(on)2}$			
2N6788, 2N6788U		I_D = rated I_{D1} (see 1.3)			.35	Ω
2N6790, 2N6790U					.85	Ω
2N6792, 2N6792U					1.90	Ω
2N6794, 2N6794U					3.10	Ω
Forward voltage (source drain diode)	4011	Condition A, $I_S = I_{D1}$	V_{SD}			
2N6788, 2N6788U					1.8	V
2N6790, 2N6790U					1.5	V
2N6792, 2N6792U					1.4	V
2N6794, 2N6794U					1.2	V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate to source voltage threshold	3403	$V_{DS} \geq V_{GS}$, $I_D = 0.25$ mA dc	$V_{GS(th)2}$	1.0		V dc
Gate current	3411	Bias condition C, $V_{GS} = \pm 20$ V dc, $V_{DS} = 0$ V dc	I_{GSS2}		± 200	nA dc
Drain current	3413	Bias condition C, $V_{GS} = 0$ V dc, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		0.25	mA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, bias condition A, pulsed, (see 4.5.1);	$r_{DS(on)3}$			
2N6788, 2N6788U		I_D = rated I_{D2}			0.54	Ω
2N6790, 2N6790U					1.50	Ω
2N6792, 2N6792U					4.00	Ω
2N6794, 2N6794U					6.60	Ω
Low temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage threshold	3403	$V_{DS} \geq V_{GS}$, $I_D = 0.25$ mA dc	$V_{GS(th)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}, V_{GS} = 10 V dc, gate drive impedance = 7.5Ω				
Turn-on delay time			$t_{d(on)}$			
2N6788, 2N6788U		V_{DD} = 35 V dc			40	ns
2N6790, 2N6790U		V_{DD} = 74 V dc			40	ns
2N6792, 2N6792U		V_{DD} = 175 V dc			40	ns
2N6794, 2N6794U		V_{DD} = 225 V dc			40	ns
Rise time			t_r			
2N6788, 2N6788U		V_{DD} = 35 V dc			70	ns
2N6790, 2N6790U		V_{DD} = 74 V dc			50	ns
2N6792, 2N6792U		V_{DD} = 175 V dc			35	ns
2N6794, 2N6794U		V_{DD} = 225 V dc			30	ns
Turn-off delay time			$t_{d(off)}$			
2N6788, 2N6788U		V_{DD} = 35 V dc			40	ns
2N6790, 2N6790U		V_{DD} = 74 V dc			50	ns
2N6792, 2N6792U		V_{DD} = 175 V dc			60	ns
2N6794, 2N6794U		V_{DD} = 225 V dc			60	ns
Fall time			t_f			
2N6788, 2N6788U		V_{DD} = 35 V dc			70	ns
2N6790, 2N6790U		V_{DD} = 74 V dc			50	ns
2N6792, 2N6792U		V_{DD} = 175 V dc			35	ns
2N6794, 2N6794U		V_{DD} = 225 V dc			30	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area test	3474	See figure 10; V_{DS} = 80 percent of rated $V_{BR(DSS)}$, $V_{DS} \leq 200$ V; t_p = 10 ms				
Electrical measurements		See table I, subgroup 2				
Single pulse unclamped inductive switching 3/	3470	See 4.3.3, c = 0, 116 devices				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B	$Q_{g(on)}$			
On-state charge						
2N6788, 2N6788U					18.0	nC
2N6790, 2N6790U					14.3	nC
2N6792, 2N6792U					22.0	nC
2N6794, 2N6794U					25.0	nC
Gate to source charge			Q_{gs}			
2N6788, 2N6788U					4.0	nC
2N6790, 2N6790U					3.0	nC
2N6792, 2N6792U					3.0	nC
2N6794, 2N6794U					6.0	nC
Gate to drain charge			Q_{gd}			
2N6788, 2N6788U					9.0	nC
2N6790, 2N6790U					9.0	nC
2N6792, 2N6792U					14.0	nC
2N6794, 2N6794U					18.0	nC
Reverse recovery time	3473	Condition A, di/dt = 100 A/µs, $V_{DD} \leq 50$ V, $I_D = I_{D1}$	t_{rr}			
2N6788, 2N6788U					240	ns
2N6790, 2N6790U					400	ns
2N6792, 2N6792U					650	ns
2N6794, 2N6794U					900	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.
3/ This test need not be performed in group A when performed as a 100-percent screen.

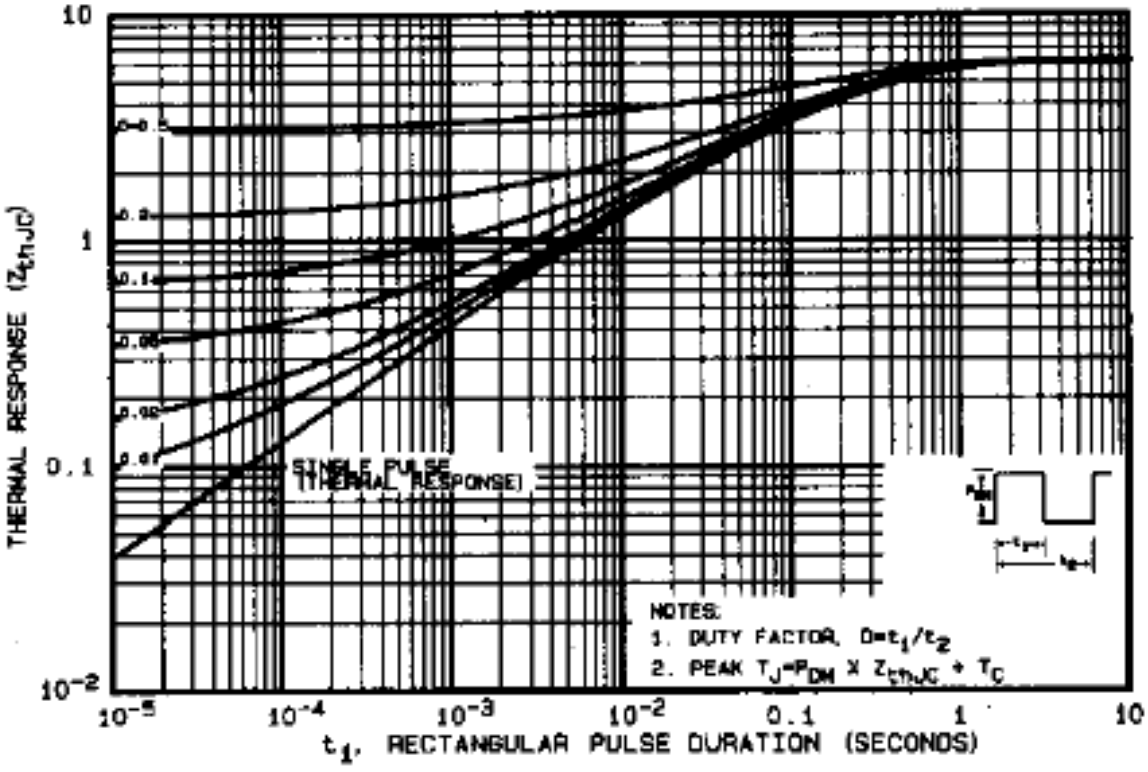
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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection 1/	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycle	1051	Condition G, 500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2	
Subgroup 2 2/			45 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours, V_{GS} = 80 percent of rated (see 1.3)	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices c = 0
Barometric pressure (reduced)	1001	Test condition C, V_{ISO} = V_{DS}, $I_{(ISO)}$ = .25 mA (max)	
2N6792, 2N6792U 2N6794, 2N6794U		V_{DS} = 300 V V_{DS} = 300 V	
Subgroup 10			
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer	22 devices c = 0

1/ JANHC and JANKC devices are qualified in accordance with MIL-PRF-19500.
2/ A separate sample may be pulled for each test.

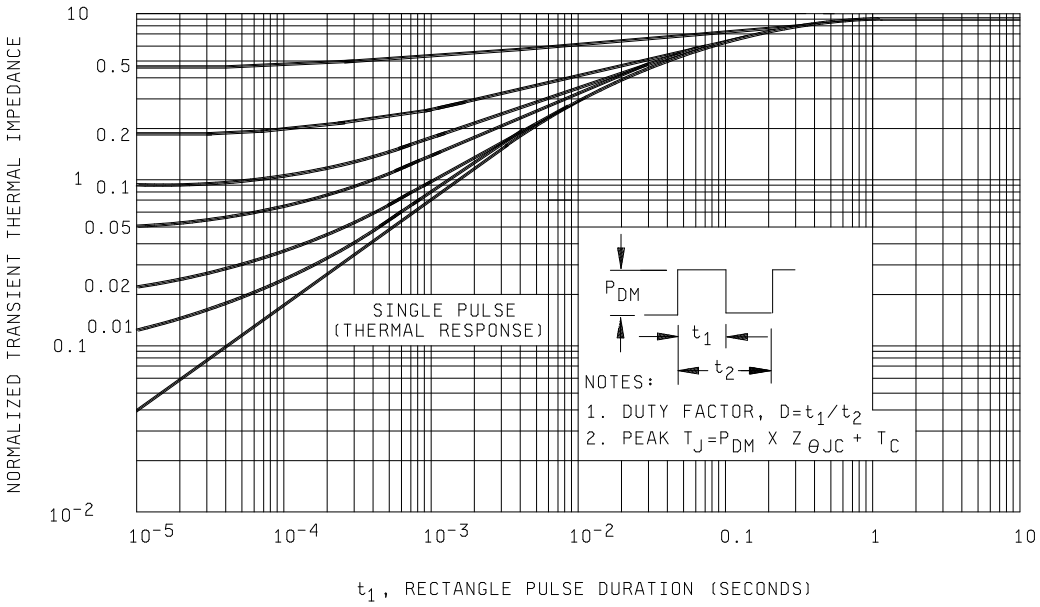
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2N6788, 2N6790, 2N6792, and 2N6794 only

FIGURE 8. Thermal impedance curves.

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2N6788U, 2N6790U, 2N6792U, and 2N6794U only

FIGURE 8. Thermal impedance curves - Continued.

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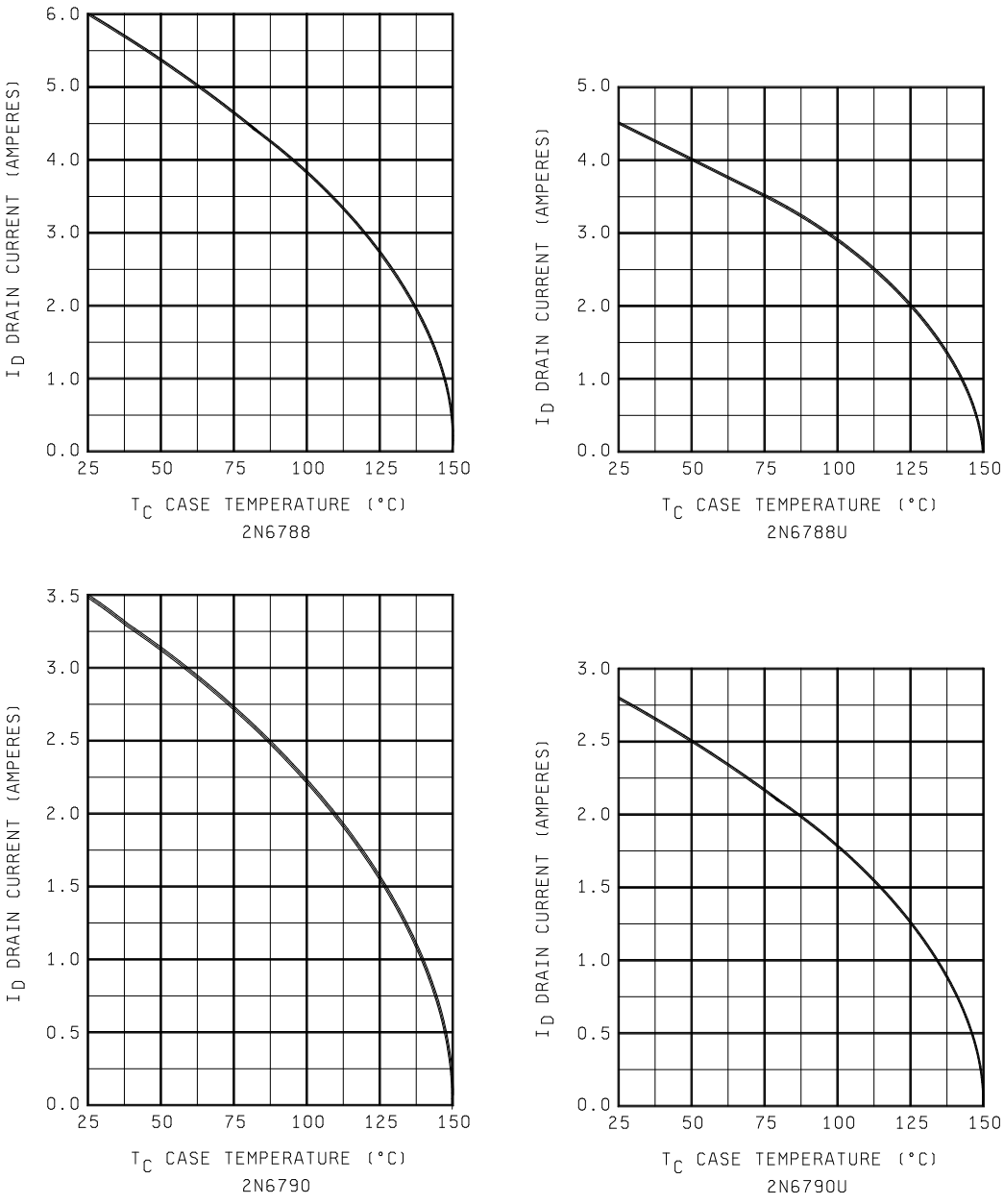
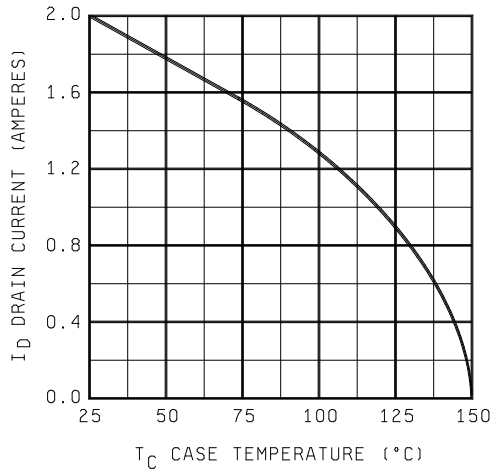
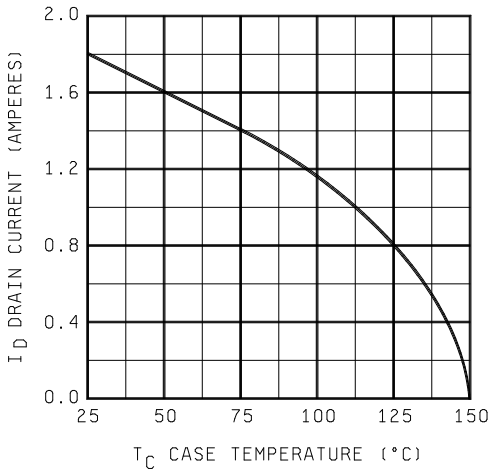


FIGURE 9. Maximum drain current versus case temperature graphs.

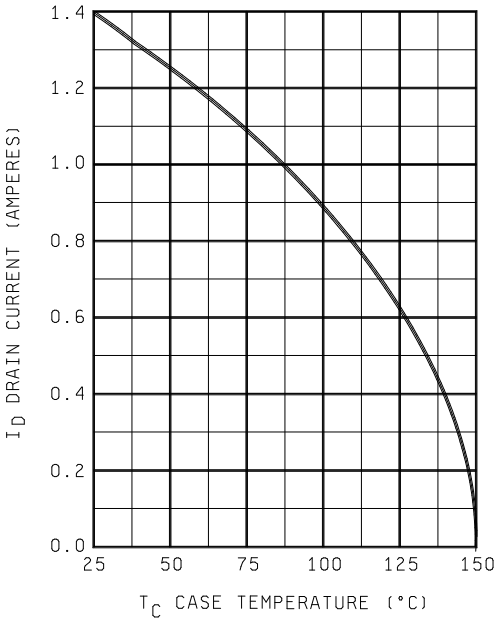
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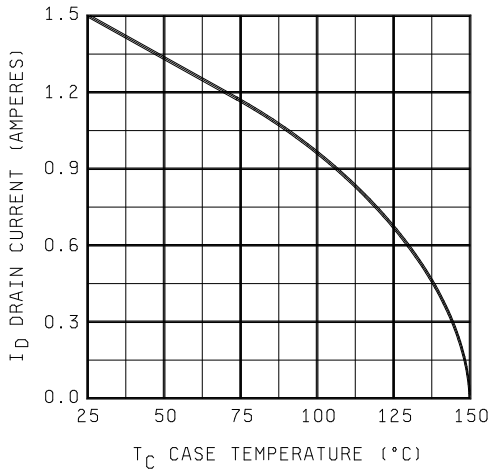
2N6792



2N6792U



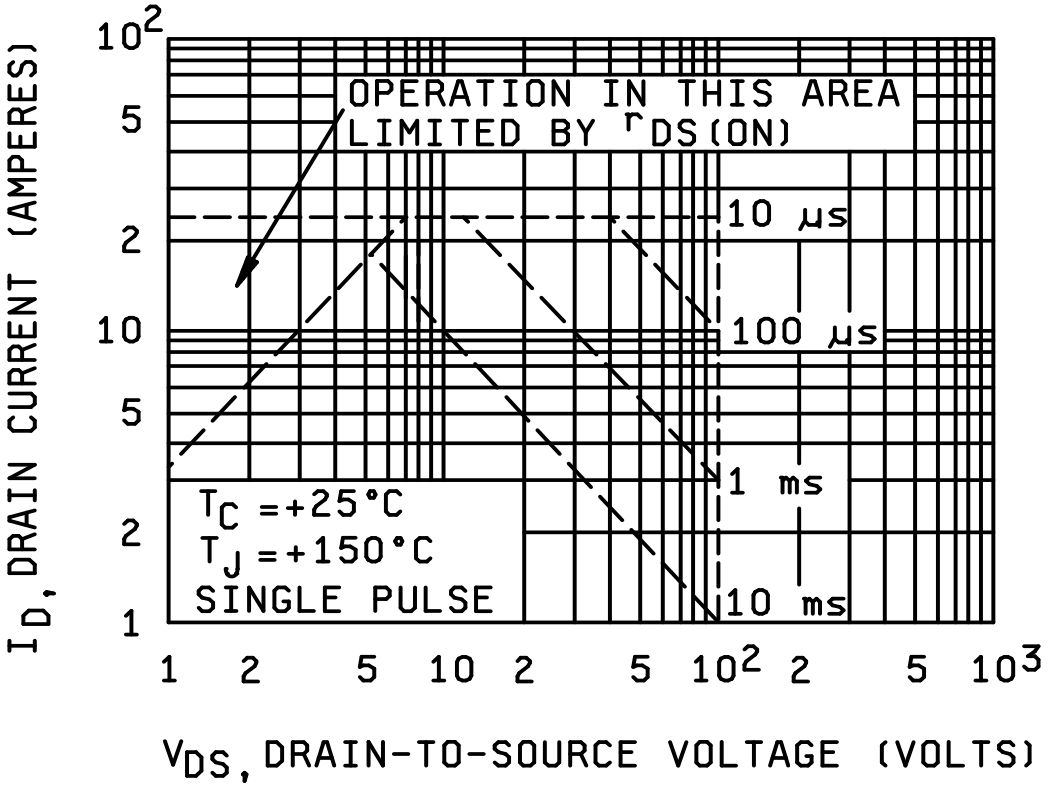
2N6794U



2N6794

* FIGURE 9. Maximum drain current versus case temperature graphs - Continued.

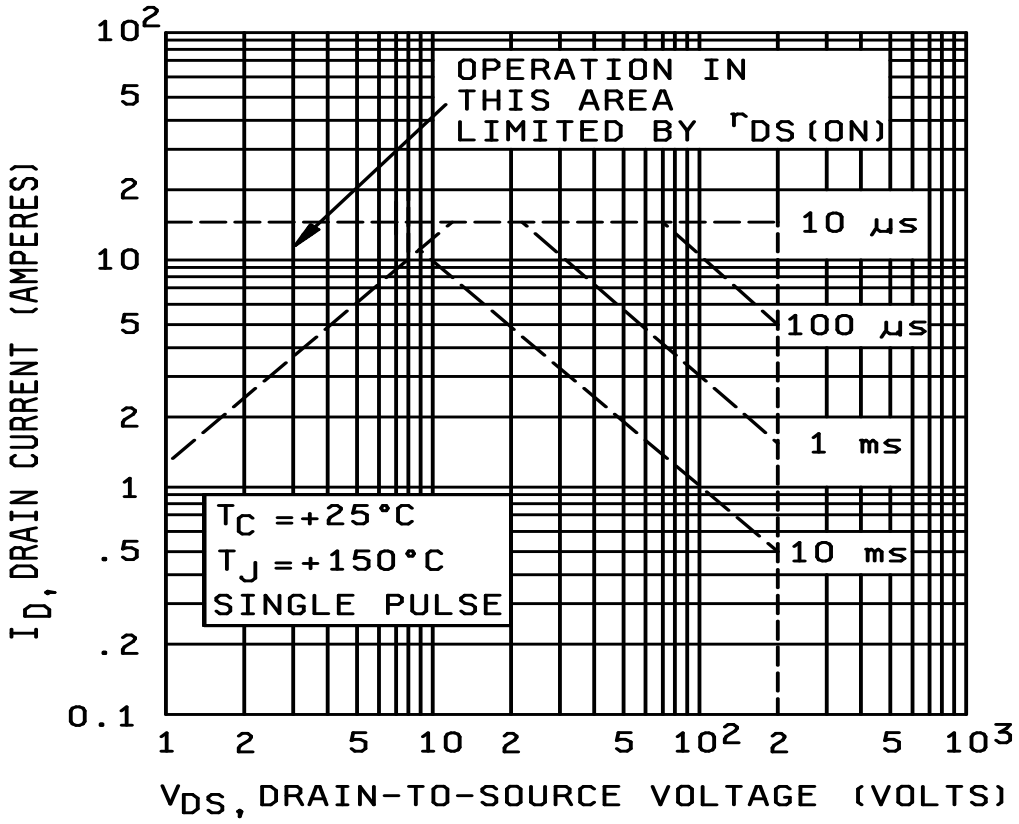
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2N6788, 2N6788U

FIGURE 10. Maximum safe operating area.

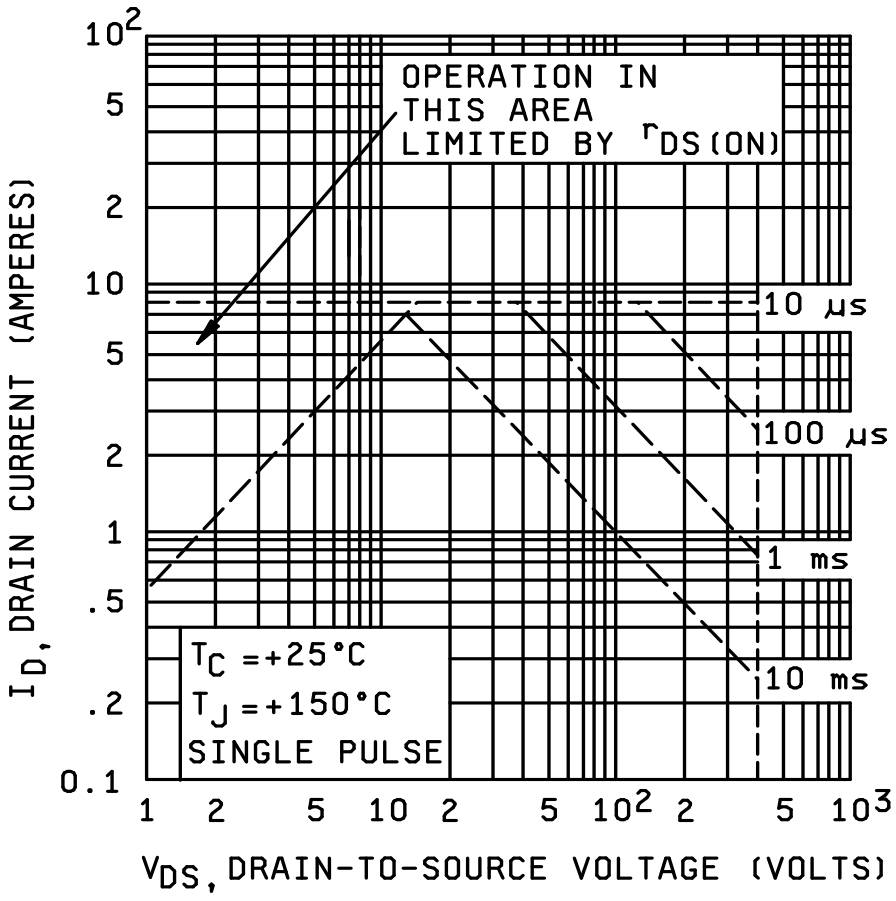
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2N6790, 2N6790U

FIGURE 10. Maximum safe operating area - Continued.

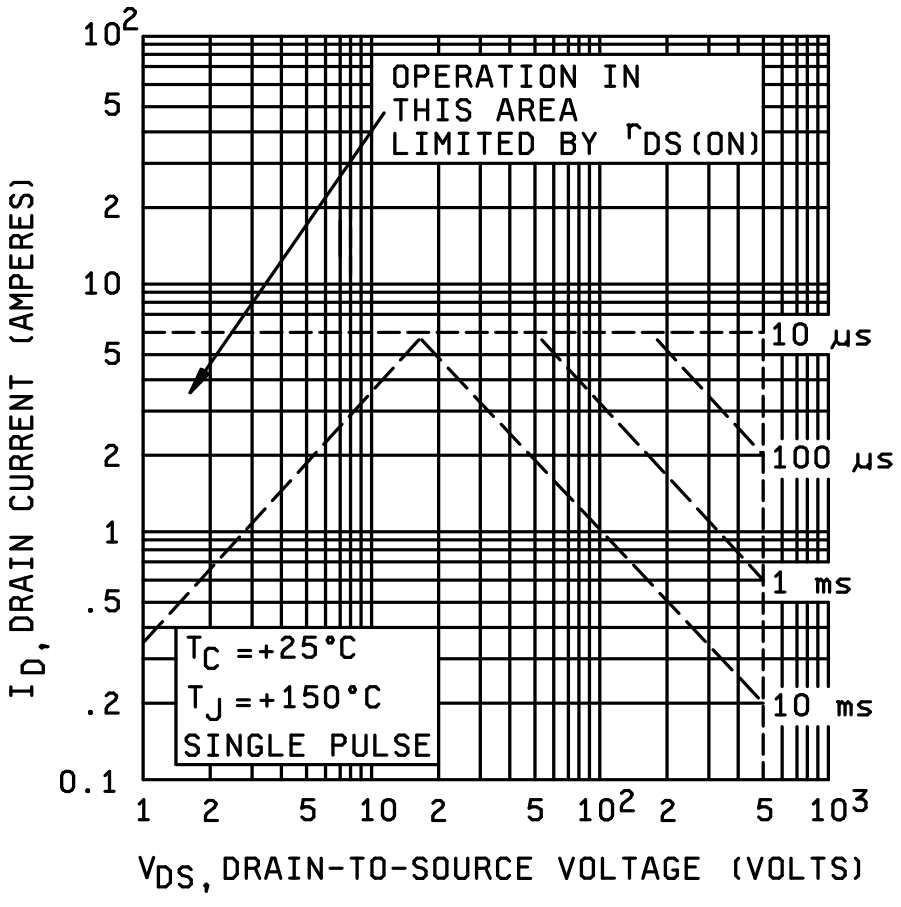
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2N6792, 2N6792U

FIGURE 10. Maximum safe operating area - Continued.

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2N6794, 2N6794U

FIGURE 10. Maximum safe operating area - Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.6.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 3, 4, 5, 6, and 7).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Cross-reference complement list. Parts from this specification may be used to supersede the following commercial Part or Identifying Number (PIN) listed below. Complementary transistors are covered by MIL-PRF-19500/564:

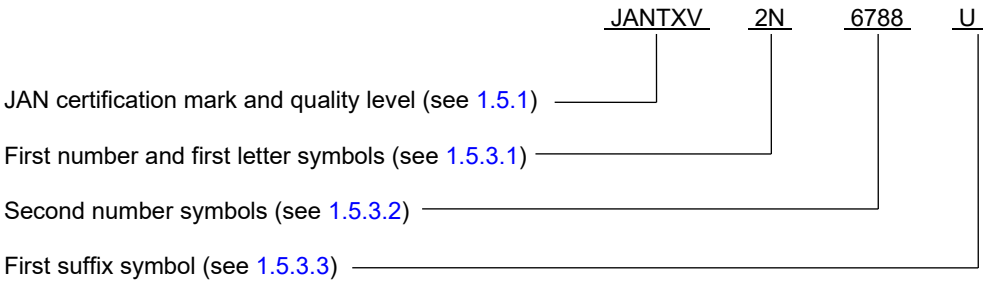
Preferred types (1)	Commercial types
2N6788	IRFF120
2N6790	IRFF220
2N6792	IRFF320
2N6794	IRFF420
2N6788U	IRFE120
2N6790U	IRFE220
2N6792U	IRFE320
2N6794U	IRFE420

(1) Prefixes are JAN, JANTX, JANTXV, or JANS

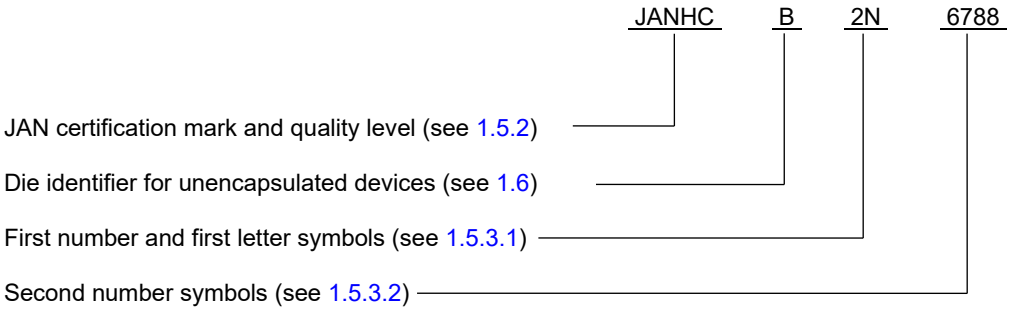
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6.5 PIN construction example.

6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.6 List of PINs.

6.6.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N6788	JANTX2N6788	JANTXV2N6788	JANS2N6788
JAN2N6788U	JANTX2N6788U	JANTXV2N6788U	JANS2N6788U
JAN2N6790	JANTX2N6790	JANTXV2N6790	JANS2N6790
JAN2N6790U	JANTX2N6790U	JANTXV2N6790U	JANS2N6790U
JAN2N6792	JANTX2N6792	JANTXV2N6792	JANS2N6792
JAN2N6792U	JANTX2N6792U	JANTXV2N6792U	JANS2N6792U
JAN2N6794	JANTX2N6794	JANTXV2N6794	JANS2N6794
JAN2N6794U	JANTX2N6794U	JANTXV2N6794U	JANS2N6794U

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* 6.6.2 List of PINs for unencapsulated devices. The following is a list of possible PINs for unencapsulated devices available on this specification sheet. The qualified die suppliers with the applicable letter version (example, JANHCA2N6788) will be identified on the QML.

JANC ordering information			
PIN	Manufacturer		
	69210	17856	43611
2N6788	JANHCA2N6788 JANKCA2N6788	JANHCC2N6788 JANKCC2N6788	JANHCE2N6788 JANKCE2N6788
2N6790	JANHCA2N6790 JANKCA2N6790	JANHCC2N6790 JANKCC2N6790	JANHCF2N6790 JANKCF2N6790
2N6792	JANHCA2N6792 JANKCA2N6792	JANHCD2N6792 JANKCD2N6792	
2N6794	JANHCA2N6794 JANKCA2N6794	JANHCD2N6794 JANKCD2N6794	

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
* Air Force - 19, 71

Preparing activity:
DLA - CC

(Project 5961-2021-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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